

PETER K BABY

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PROFESSIONAL SUMMARY

Junior Software Engineer with 2 years of full-stack development experience building scalable web applications and AI/LLM-powered solutions. Experienced in Python, FastAPI, REST APIs, PostgreSQL, React, TypeScript, Pydantic, prompt engineering, AI agents, structured outputs, tool integration, JWT authentication, RBAC, Docker, Kafka, and Google Cloud Run. Built maintainable backend services and reusable components by translating product requirements into production-ready APIs, validation logic, and user-focused applications.

WORK EXPERIENCE

Junior Software Engineer

Jan 2025 - Present

Bitflies Evolve Pvt Ltd

Kozhikode

- Contributed to 2 production products, SENSY Fleet Management System and KERAA Property & Lease Management System, as a full-stack developer across backend APIs, dashboards, and deployment workflows.
- Developed RESTful APIs with FastAPI and PostgreSQL for user management, authentication workflows, lease workflows, fleet records, and role-based access across production applications.
- Developed LLM-powered features using LangChain, Retrieval-Augmented Generation (RAG), OpenAI API, and Google Gemini API, applying prompt engineering and context-aware workflows for AI-powered applications.
- Built modular React and TypeScript components with Redux-based state management, reusable business logic, protected routes, and API integration patterns.
- Implemented form validation, conditional rendering, asynchronous data handling with Axios, and responsive Tailwind CSS interfaces for operational dashboards and data-entry workflows.
- Integrated Google Maps APIs for live geolocation tracking, route generation, route optimization, and spatial data visualization in fleet workflows.
- Secured application modules with JWT authentication and Role-Based Access Control (RBAC), supporting separate access flows for administrators, staff, and users.
- Containerized backend services with Docker and deployed applications on Google Cloud Run, applying cloud deployment and CI/CD practices.
- Implemented OAuth 2.0 authentication and authorization flows for secure integration with external services, alongside Pydantic models and schema-based validation to define typed request/response contracts and keep backend behaviour predictable and testable.
- Collaborated through Git, GitHub/GitLab, structured code reviews, Postman, and Insomnia to validate APIs, resolve issues, and prepare releases.

Python Full-Stack Developer - Intern

Sept 2024 - Dec 2024

Bitflies Evolve Pvt Ltd

Kozhikode

- Assisted on internal web application modules at Bitflies Evolve Pvt Ltd, gaining industry experience in FastAPI, React, PostgreSQL, authentication, and API documentation.
- Built backend REST APIs using Python and FastAPI to support authentication, data processing, and frontend integration for full-stack workflows.
- Created reusable React components for form-driven workflows and dashboard interfaces, improving consistency across frontend screens.

- Applied authentication mechanisms and role-based access control to secure user access by role and module.
- Documented and tested API endpoints using OpenAPI and Postman, improving handoff between backend and frontend development.

Python Full-Stack Developer - Intern

Dec 2023 - June 2024

Mashupstack

Trivandrum

- Completed a 6-month full-stack development internship focused on Django, React, MySQL, REST API development, database design, and frontend integration.
- Delivered full-stack web application modules using Django, React, and MySQL, covering REST APIs, database operations, and frontend user interfaces.
- Designed RESTful API endpoints and configured MySQL schemas for authentication, content, and transactional workflows.
- Practiced Git-based development, API testing, debugging, and component-based frontend development through guided project work.

PROJECTS

SENSY

2025

Fleet Management System

- Designed and implemented a fleet tracking system using FastAPI, PostgreSQL, and React to manage vehicle records, trip data, and tracking workflows.
- Enabled map-based route generation, live geolocation tracking, and route optimization using Google Maps APIs.
- Technologies: React, Redux, Tailwind, Python FastAPI, PostgreSQL

KERAA

2025

Property & Lease Management System

- Built a property and lease management platform using FastAPI and React to handle property records, lease workflows, installments, and user access.
- Automated lease and installment workflows to reduce manual tracking effort for recurring property operations.
- Designed role-based dashboards for administrators, staff, and users.
- Technologies: React, Redux, PrimeReact, FastAPI

OTT Platform

2024

Streaming Web Application

- Engineered a subscription-based OTT platform using Django, React, and MySQL with secure authentication and content management features.
- Integrated JWT authentication, REST APIs, and database models for user subscriptions and streaming content.
- Improved frontend responsiveness through reusable React components and optimized API-driven views.
- Technologies: Django, DRF, React, MySQL

SKILLS

Programming Languages: Python, JavaScript, TypeScript

Backend Technologies: FastAPI, Django, REST API Development, Async, Pydantic, JWT Authentication, role-based access control (RBAC), Webhooks, OAuth 2.0

Frontend Technologies: React, Next.js, TypeScript, Tailwind CSS, HTML, CSS
Databases: PostgreSQL, MySQL, Redis, SQLAlchemy, Alembic
AI & LLM: AI Agents, LLM Integration, LangChain, RAG, OpenAI API, Google Gemini API, Prompt Engineering, Tool Calling, Structured Outputs, Context Management, Guardrails, Schema Validation
Tools & Platforms: Git, GitHub, GitLab, Pytest, Postman API Client, Insomnia API Client
Libraries & Frameworks: LangChain, Redux, NumPy, Pandas, BeautifulSoup
Cloud & DevOps: GCP, Cloud Run, CI/CD, Docker, SonarQube

Core Concepts: Agile, API Design, Microservices, Backend Development, Authentication Systems, Event-Driven Architecture, CI/CD, Full-Stack Development

CERTIFICATIONS AND ACHIEVEMENTS

Certifications	Python for Data Science, AI and Development by IBM, offered through Coursera Python Project for Data Science by IBM, offered through Coursera Django 3 - Full Stack Websites with Python Web Development by Nick Walter through Udemy HackerRank - Python (Basic)
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POSITION OF RESPONSIBILITY

Media and Outreach Head <i>YASYA 23 (Cultural Fest), St. Thomas Institute for Science and Technology</i>	Dec 2022 - May 2023
· Video Editor · Organized Media Department at College	
NSS Group Leader <i>National Service Scheme</i>	Jun 2017 - Aug 2019
· Coordinated team members on various events.	
Yasya Event Coordinator	
· Coordinated onstage and offstage events for Yasya, the College Cultural Festival, 2023.	

EDUCATION

Bachelor of Technology in Computer Science Engineering St. Thomas Institute For Science and Technology, Trivandrum APJ Abdul Kalam Technological University, Kerala	Aug 2019 - June 2023
HSE, Class XII St. John’s Model HSS, Trivandrum Kerala State Board	2017 - 2019
CBSE, Class X Infant Jesus Central School, Trivandrum CBSE	2017

LANGUAGES

English, Malayalam